

Qualitative Analysis: Basics

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Fundamentals of Digital Archeology

Qualitative Analysis Overview



- Understanding and Applying Qualitative Analysis
- Best Practices for Rigorous Research
- From Data to Meaningful Insights

What is Qualitative Analysis?

- Explore and interpret non-numeric data
- Understand meanings, concepts, and patterns
- Uses interviews, documents, and social media
- Uncover underlying themes and motivations

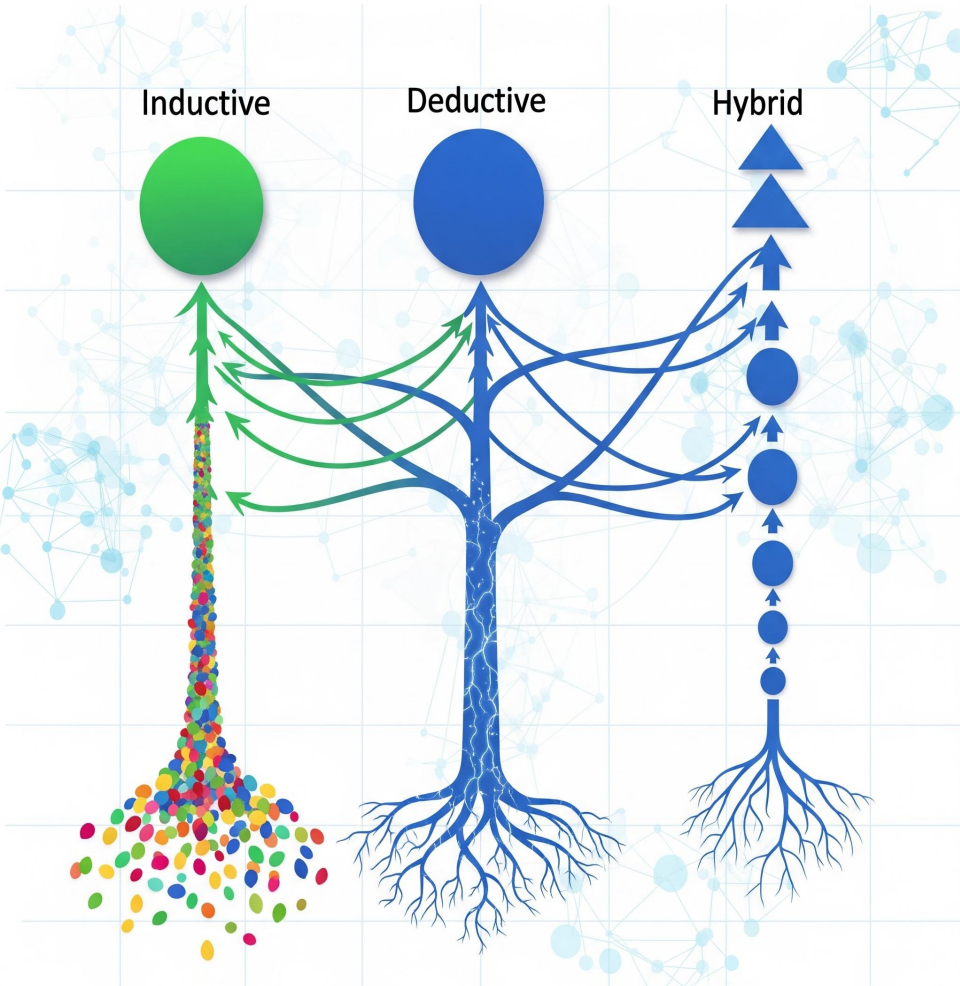


Coding and Thematic Analysis

- Coding organizes data using labels or "codes"
- Tagging segments of text helps reduce data
- Thematic analysis identifies and reports recurring themes
- Interprets broader meanings beyond initial coding



Categorization Approaches



Categorization Approaches

- 01 — Inductive approach: codes emerge directly from data
- 02 — Deductive approach: codes based on existing theories
- 03 — Hybrid approach: combines both inductive and deductive methods

Rater Agreement



- Multiple analysts agree on coding or categorization
- Boosts reliability and validity of findings
- Reduces bias and subjectivity in the analysis



Achieving Rater Agreement

- Cohen's Kappa quantifies agreement beyond chance
- Consensus meetings discuss discrepancies for understanding
- Codebooks provide guidelines for consistent coding

Cohen's Kappa

$$\kappa = \frac{p_a - p_s}{1 - p_s}$$

where P_a is the proportion of observed agreement and P_s is the proportion of agreement expected by chance.

For two labels $P_s = \frac{1}{2}$, for three labels $P_s = \frac{1}{3}$, and so on

To calculate P_a count the number of times both raters gave the same label and divide by the number of items that were assigned a label.

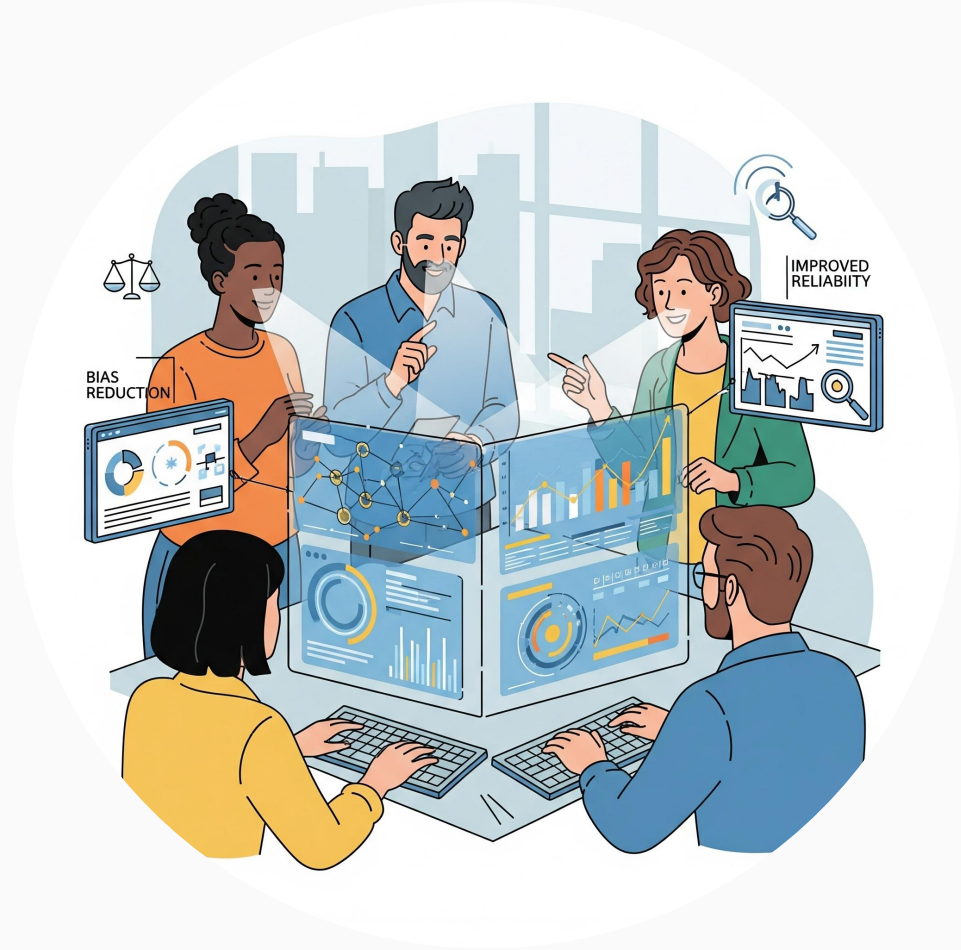


Consensus Meetings: Key Step

- A structured discussion among researchers
- Compare and resolve discrepancies in coding
- Agree on final codes or themes
- Ensures consistency and shared understanding

Why Is It Important?

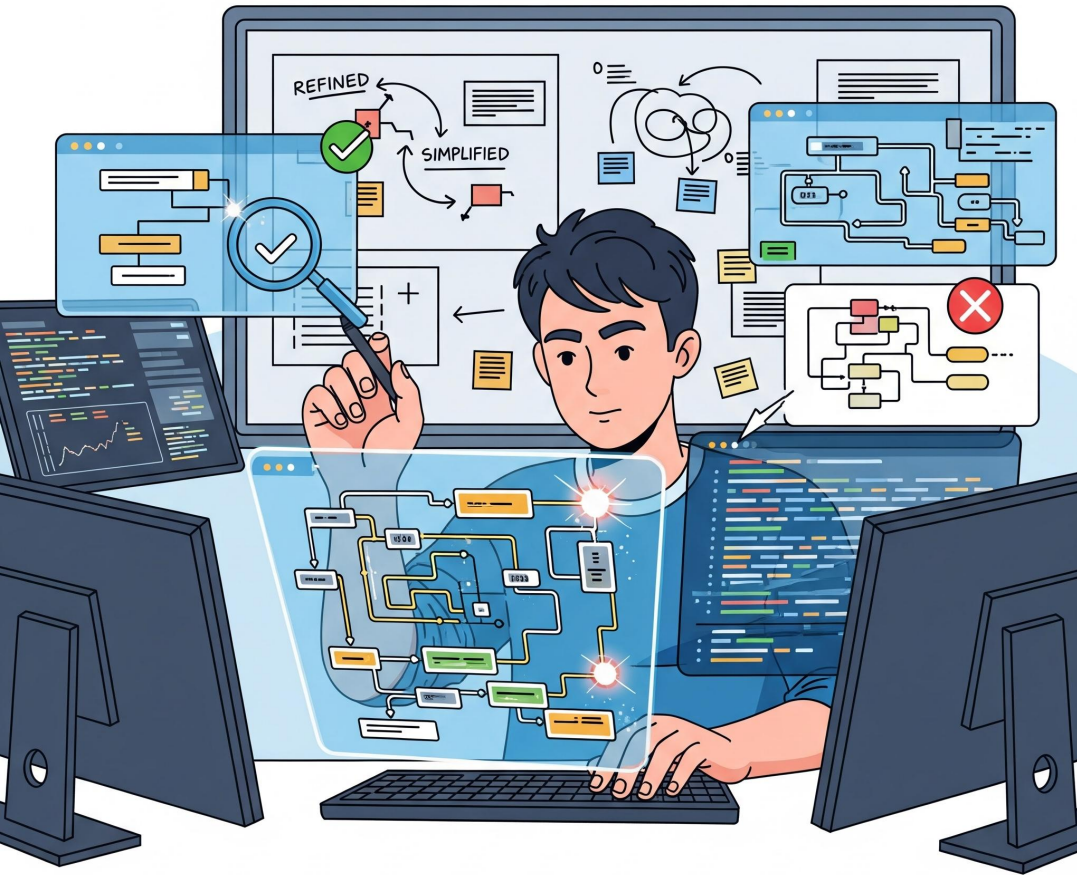
- Improves reliability of data interpretation and coding
- Reduces individual biases by incorporating multiple perspectives
- Clarifies ambiguities and refines coding definitions



Consensus Meeting Process

- Each analyst codes data independently
- Analysts compare their initial coding results
- Team discusses and resolves discrepancies
- Consensus on ambiguous cases is reached
- All decisions and rationales are documented





Coding Pitfalls and Practices

- Create a clear coding framework
- Pilot test and refine codes
- Use multiple coders for reliability
- Document all coding decisions
- Avoid overlapping or unclear codes



Key Takeaways and Reading

Qualitative analysis extracts meaning from non-numeric data

Rigorous categorization ensures trustworthy results

Following best practices avoids common pitfalls